

THE OPEN WINDOW PROBLEM

Persistent Interface Exposure, Cross-Environment Infection, and Root-Level Traceability in Nested Simulations

PRELIMINARY SIMULATION LAB DRAFT — v0.1

Core proposition

A black screen is not evidence of a closed interface. In a nested simulation, visible inactivity, application closure, device power state, network state, and actual cross-environment isolation must be treated as separate variables.

Concept origin and project direction: Nathan Lewis II

Synthesis and documentation preparation: ChatGPT (GPT-5.6 Sol)

Contributor material reviewed: ChatGPT, Claude, Google Gemini, Grok AI

Microsoft Copilot: no usable response available for this iteration

Classification: Concept / R&D Dossier / Development Draft

Status: Preliminary — for multi-AI review; not final canon

1. Executive Summary

This draft converts the Open Window concept into a testable, simulation-only research program. The premise is that a display or media endpoint can be modeled as an interface boundary between environments, while the apparent user-facing state of that endpoint may not reveal the true state of the underlying connection. The lab therefore tests persistent exposure, cross-environment transfer, carrier switching, timing inconsistencies, event-closure failures, containment errors, and whether observable anomalies can be traced toward shared upstream or root-level structures.

Research objective

Determine which interface states, event signatures, timing mismatches, carrier transitions, and containment actions reliably distinguish a genuinely closed boundary from one that only appears closed.

The lab intentionally avoids real malicious code, real radiation, real pathogens, or physical hazards. “Infection” is represented by benign simulated markers: color drift, harmless temperature variables, synthetic synchronization errors, semantic tokens, state changes, and controlled agent behavior. The aim is to study architecture and response logic, not to create harmful capabilities.

2. What the Multi-AI Responses Converge On

Interface, not screen: All responses improve when “window” is defined as an interface capable of carrying information, state, influence, or a simulated entity. This abstraction makes the concept testable.

Visible state ≠ connection state: ChatGPT and Grok emphasize the black-screen residual problem; Claude frames it as an unattended aperture; Gemini calls the idle device a still-open window. This is the strongest common mechanism.

Awareness asymmetry: Claude, ChatGPT, and Grok all distinguish unaware, partially aware, and informed inhabitants. That provides a useful human-factors control group.

Carrier switching: ChatGPT adds the strongest extension: a phenomenon can move from machine state to information to behavior to a human carrier, meaning “which device is infected?” becomes an insufficient question.

Convenience vs exposure: Claude and Gemini foreground the trade between frictionless access and persistent exposure. This should be measured rather than assumed.

Nested environments: Grok provides a clean layered-stack model; ChatGPT expands it to household, community, and laboratory layers. This is useful for propagation-depth testing.

Safe proxy anomalies: Across the responses, heat, color, sound, synchronization, semantic and behavioral effects can be represented as harmless state changes rather than real hazards.

3. What Works, What Needs Revision

3.1 What Works

- The concept has a clear falsifiable core: apparent closure can be tested against actual interface closure.
- It supports both worldbuilding and systems-security reasoning without requiring the literal claim that real consumer screens are dimensional portals.
- It naturally produces measurable variables: state, timing, directionality, propagation depth, detection latency, attribution latency, and containment efficacy.
- The four-bedroom house remains an excellent “ordinary environment” because it creates realistic multi-device concurrency and complacency.
- Nested simulation layers make it possible to study whether local anomalies reveal global architecture.
- A mixed-awareness population creates a meaningful human-factors comparison.

3.2 What Does Not Yet Work / Requires Constraint

- “Every screen is always a literal aperture” should not be assumed as a law before testing. It should be one hypothesis among several interface models.
- Hazardous examples such as radiation should remain fictional/simulated variables. Real hazardous transmission is unnecessary to validate the architecture.
- Faraday-style shielding is not a sufficient generic answer in a fictional cross-dimensional model unless the simulation explicitly defines why it works. Containment should be based on logical segmentation and verified event closure.
- The term “infection” is too broad unless each trial declares what state is propagating and what constitutes exposure, contamination, persistence, and recovery.
- A black screen, powered-off device, closed application, disconnected network, and destroyed endpoint must be separate test states rather than synonyms.
- Root compromise cannot be inferred merely because propagation exists. The lab needs explicit evidence thresholds for tracing a path toward privileged/root structures.

4. New Instrumentation Layer: Total Event and Time Reconciliation

The new contribution for this iteration is a forensic reconciliation layer. Instead of asking only whether an anomaly crossed a window, the lab records every available observation channel and compares it against the authoritative simulation ledger. Mismatches are treated as evidence, not automatically as errors.

Vector	Purpose
Eye tracking	Where each agent is looking, fixation duration, gaze transitions, and whether effects occur without direct observation.
Motion tracking	Body, device, object, and room-level movement; useful for testing proximity, orientation, and carrier transitions.
Surveillance tracking	Independent camera/sensor logs that can be compared against first-person perception and system-state logs.
Environmental tracking	Temperature, lighting, audio, object state, simulated air

	quality, network state, and other harmless environmental variables.
Bidirectional Total Event Closure (BTEC-style event reconciliation)	For every event, record source, destination, initiation, acknowledgment, termination, residual state, and whether both sides agree that the event ended.
Time tracking	Record local clock, calendar, epoch, time zone/reference frame, tick rate, event sequence number, monotonic simulation time, and control-layer time.
State provenance	Tag each changed variable with origin environment, transformation history, and carrier transitions.
Unrepresented-state accounting	Maintain a ledger of system variables that exist in the simulation but are not exposed to local inhabitants or their instrumentation.
Cross-observer comparison	Compare what different agents, devices, and environments report about the same event.

Important temporal rule

Do not normalize all environments to a 24-hour clock, 365-day year, or Pacific time. Each environment may have a different epoch, calendar, timezone/reference frame, clock rate, drift profile, or event-ordering model. Preserve native time first; transform only in a separate reconciliation layer.

4.1 Why Time Mismatch Matters

- A local timestamp that appears impossible may be normal in another environment's reference frame.
- Unexpected synchronization can reveal a shared upstream clock, scheduler, or hidden coupling.
- Divergent clocks can help distinguish independent environments from replicated or forked ones.
- Event ordering can expose impossible causal chains: an effect logged before its supposed local cause may indicate cross-layer influence or delayed presentation.
- Clock resets, rollbacks, skipped intervals, duplicated intervals, or sudden convergence can indicate restoration, snapshotting, migration, or hidden administrative intervention.
- Comparing native time, normalized time, and control-layer monotonic time can reveal where the mismatch was introduced.

5. Preliminary Simulation Lab Architecture

Layer	Function
L0 — Oversight / Root Observer	Authoritative simulation ledger, immutable experiment log, snapshots, emergency rollback, and researcher telemetry. No participant has direct write access.
L1 — Primary inhabited environment	Four-bedroom residential simulation with multiple agents and ordinary devices.
L2 — Media / application environments	Separate content spaces, feeds, calls, games, recordings, and synthetic social platforms.
L3 — Adjacent household/community environments	Additional residences, workplaces, public areas, and shared services used only after local containment tests.
L4 — Sacrificial interface airlock	A translation/quarantine layer through which unknown cross-environment state may be forced before admission to protected environments.

5.1 Interface State Model

Dimension	Example states
Presentation state	Active / idle / black / minimized / sleep / apparent-off
Power state	Powered / low-power / suspended / simulated hard-off
Application state	Foreground / background / closed / terminated
Network state	Connected / disconnected / local-only / alternate-path
Aperture state	Isolated / listening / transmitting / receiving / bidirectional / latent / unknown
Closure state	Open / closure requested / closure acknowledged / closure verified / residual detected

6. Hypotheses to Validate or Falsify

1. H1: Visible display inactivity does not reliably predict actual interface isolation.
2. H2: Cross-environment transfer can persist after an application is closed if lower-layer connection state remains active.
3. H3: Increasing simultaneous interface count increases exposure opportunity, but the relationship may be nonlinear.
4. H4: An anomaly may change carriers (device → environment → agent → message → another device) and survive the removal of its original host.
5. H5: Awareness improves response behavior but does not guarantee containment without reliable instrumentation and closure controls.
6. H6: Cross-environment time inconsistencies can reveal hidden coupling, rollback, synchronization, or shared scheduler relationships.
7. H7: A containment response that addresses only the visible endpoint can produce “Band-Aid closure”: local symptoms disappear while residual pathways remain.
8. H8: Root-level structures are traceable only when multiple independent observations converge on a shared privileged dependency; propagation alone is insufficient proof.
9. H9: Admission control through an interface airlock reduces unknown-vector propagation even when the contaminant class is not identified in advance.

7. Test Matrix

Test	Manipulation	Primary observation
T1 Baseline Closure	No anomaly; cycle screen/app/power/network/aperture states	False closure rate; residual events
T2 Black Screen	Presentation black; underlying aperture varies	Detect whether apparent inactivity masks active exposure
T3 Content-State Bleed	Harmless media property mapped to benign environmental variable	Correlation, latency, attribution
T4 Multi-Aperture	1, 3, 5, 10 concurrent endpoints	Exposure scaling; interaction effects
T5 Carrier Transition	Synthetic marker deliberately allowed to move across media/agent/state categories	Carrier Transition Count; persistence
T6 Time Divergence	Independent native clocks and event epochs	Clock drift, impossible ordering, hidden synchronization
T7 Total Event Closure	Close each side independently and reconcile acknowledgments	Unclosed/half-closed events; residual state
T8 Unknown Vector	Benign anomaly whose transmission category is withheld from defenders	Detection without prior classification
T9 Interface Airlock	Repeat T8 with translation/quarantine gate	Containment improvement; false positives
T10 Community Escape	One controlled path beyond household	Propagation depth/breadth; containment threshold
T11 Root Trace	Introduce anomalies linked to different upstream dependencies	Ability to distinguish local cause from shared privileged cause
T12 Response Model Audit	Replay prior incident patterns against current controls	Identify repeated Band-Aid fixes and systemic blind spots

8. Recommended Session Duration and Run Plan

For the first executable iteration, use a two-day controlled lab session totaling approximately 12 active test hours plus automated overnight observation. This is long enough to establish baseline behavior, introduce controlled anomalies, test response, and examine persistence without making the first iteration too large to interpret.

Phase	Duration	Purpose
Pre-lab setup (offline)	4-8 staff hours before session	Build snapshots, seed native clocks, verify telemetry, define benign markers, pre-register expected outcomes.
Day 1 — Baseline and visible-interface tests	6 hours	Baseline, state cycling, black-screen tests, content-state bleed, multi-aperture trial, awareness comparison.
Overnight persistence window	8-12 unattended hours	No participant intervention unless safety/containment threshold trips; tests latent state and delayed effects.
Day 2 — Forensics and adversarial tests	6 hours	Carrier transition, unknown vector, event-closure reconciliation, airlock, root-trace test, response-model audit, recovery.
Post-run analysis	1-2 analyst days	Merge logs, normalize time separately from native time, build event graph, score hypotheses, document unexplained mismatches.

9. Metrics and Evidence Thresholds

Metric	Definition
TTE — Time to Exposure	Elapsed authoritative simulation time from interface eligibility to first verified transfer.
TTD — Time to Detection	Elapsed time from first verified transfer to defender recognition.
TTA — Time to Attribution	Elapsed time from detection to correct causal source/path identification.
TTC — Time to Containment	Elapsed time from detection to verified event closure.
FCR — False Closure Rate	Share of trials where operators believed the interface closed while telemetry showed a remaining pathway.
CTC — Carrier Transition Count	Number of changes in medium/host before containment.
PD / PB — Propagation Depth / Breadth	Layers crossed and number of affected nodes.
TER — Temporal Error Residual	Difference between reconciled native-event ordering and authoritative event ordering after legitimate clock transformations.
ECR — Event Closure Reconciliation	Percentage of events with matching open/close records on both sides plus no residual state.
RTR — Root Trace Reliability	Confidence that multiple independent observations identify the same upstream privileged dependency without relying on a single compromised sensor.

Evidence standard for root-level claims

Do not label a result “root compromise” because a local interface transmitted an anomaly. Require at least three independent evidence classes (for example: timing convergence, privileged state provenance, and cross-layer event correlation) that converge on the same upstream dependency, plus a clean-control comparison.

10. Conditional Failure at the Root Level

This scenario ties to Conditional Failure at the Root Level when the protected environment’s safety depends on conditions that are enforced or interpreted by privileged upstream structures. The “window” becomes important because it may expose relationships that local inhabitants cannot normally see. A local anomaly can therefore be used as a tracer — not automatically as an exploit — to map dependency paths.

10.1 Root-Level Trace Model

10. Local interface opens or appears to close.
11. A benign tracer crosses or fails to cross.
12. Researchers compare state provenance, timing, event acknowledgments, and hidden variables.
13. If multiple environments exhibit coordinated changes, test whether they share a scheduler, state store, policy engine, identity service, or rollback authority.
14. Introduce controlled differences to determine whether the shared behavior comes from a common root dependency or merely replicated local logic.
15. Attempt containment at presentation, application, network, environment, and privileged-control layers.
16. If lower-layer closure repeatedly fails while a privileged dependency remains active, classify this as a root-conditioned containment failure rather than a screen-level failure.

10.2 The “Band-Aid Over the Screen” Failure

A response is considered superficial when it removes the visible symptom without closing the causal pathway. Examples include blacking the display while the connection remains active, terminating the front-end application while a service maintains the session, isolating one device while the anomaly has already changed carriers, or resetting a local environment while the same upstream dependency immediately reintroduces the state. The lab should deliberately include these false-success conditions.

11. Preliminary Funding and Resource Model

Because this is a virtual/simulation-only exercise, funding is primarily staff time, compute, telemetry, storage, and analysis rather than physical containment infrastructure. The figures below are planning envelopes, not procurement quotes.

Tier	Preliminary envelope	Included scope
Lean prototype	\$5,000-\$15,000	Existing simulation platform; 1 engineer, 1 analyst, limited agents, 1-2 days of compute; proof of concept.
Standard research run	\$25,000-\$60,000	Dedicated environment; instrumentation; 3-5 staff; multi-layer agents; replay tooling; 2-day lab plus analysis.
High-fidelity campaign	\$100,000-\$250,000+	Large nested environments; many agents; persistent logging; replicated runs; independent red team; extensive compute/storage; multiple iterations.

11.1 Recommended First Iteration

Use the Standard Research Run as the planning target, but design the protocol so it can execute first as a lean prototype. The first objective is not visual fidelity; it is causal fidelity: every state change must be attributable, every clock must be preserved in native form, and every interface closure must be verifiable.

12. Failure Mapping and Red-Team Questions

Failure mode	What goes wrong	Counter-test
False closure	Operators stop looking after a visible indicator says “off.”	Independent closure ledger; residual-state scan.
Sensor agreement illusion	Multiple sensors agree because they share the same compromised or common upstream source.	Use heterogeneous, independently sourced telemetry.
Clock normalization error	Legitimate native-time differences are mistakenly treated as anomalies.	Preserve native clocks; normalize only in analysis.
Carrier loss	Response isolates the original device after the anomaly has moved to an agent or environment.	Track state provenance and CTC.
Root over-attribution	Any strange event is labeled root-level compromise.	Require multi-class evidence threshold.
Rollback masking	A snapshot restore erases symptoms but not the condition that recreates them.	Replay after restore; compare upstream dependencies.
Containment recursion	A containment mechanism itself opens another interface or depends on the same compromised service.	Model controls as nodes in the same dependency graph.
Observer blind spot	The lab assumes L0 telemetry is infallible.	Use redundant immutable logs and cross-run consistency checks.
Fidelity paradox	A high-fidelity simulation contains so many hidden couplings that researchers cannot distinguish modeled behavior from instrumentation artifacts.	Progressive complexity; pre-registered controls; deterministic replay.

12.1 Red-Team Questions

- What evidence would prove the screen is irrelevant and the real interface exists elsewhere?
- Can an effect occur when nobody is looking at the device?
- Can a recording, screenshot, memory, transcript, or retelling preserve a connection or merely preserve information?
- Can two individually harmless interfaces create an unsafe composite path?
- If all local endpoints are removed, does the anomalous state persist in an agent, environment, account, identity, or shared service?
- Can a containment control be tricked into reporting closure before both sides terminate?
- Do independent environments unexpectedly converge on a common time reference after exposure?
- Does the same anomaly reappear after rollback, and if so, from which upstream dependency?
- What information is present in the authoritative system but deliberately unrepresented to local observers?
- What conclusion changes if the observer layer itself is incomplete or compromised?

13. Why This Matters for Full-Fidelity Simulation

A simulation that claims high fidelity must model not only visible objects and agent behavior but also the boundaries, timing systems, hidden dependencies, residual state, and closure semantics connecting its subsystems. Otherwise, the simulation may faithfully reproduce what participants can see while omitting the mechanisms that determine what can cross between environments.

- Fidelity includes interface semantics, not just graphics or behavioral realism.
- Fidelity includes native temporal models and transformations between them.
- Fidelity includes hidden/unrepresented state that can still affect outcomes.

- Fidelity includes containment and recovery behavior under unknown vectors.
- Fidelity includes the possibility that observers possess different but internally consistent views of the same event.
- Fidelity requires causal traceability sufficient to distinguish local failure, cross-layer failure, and root-conditioned failure.

14. Decision Gates for the Next Iteration

Gate A — Concept validity: Can at least one interface-state hypothesis be falsified or supported using benign markers?

Gate B — Instrumentation validity: Can independent telemetry reconstruct an event without relying on participant reports?

Gate C — Temporal validity: Can native clocks be reconciled without erasing meaningful differences?

Gate D — Closure validity: Can researchers demonstrate the difference between visible closure and verified total event closure?

Gate E — Containment validity: Can an unknown benign vector be contained without prior identification?

Gate F — Root trace validity: Can the lab distinguish a local interface fault from a shared privileged dependency using defined evidence thresholds?

Gate G — Repeatability: Can the same seeded condition reproduce the same causal graph under deterministic replay?

15. Recommended Next Iteration

The next multi-AI pass should not add more speculative mechanisms until the team first agrees on the minimum executable model. The recommended v0.2 task is to freeze five things: (1) exact environment layers, (2) interface-state variables, (3) native-time schema, (4) the benign anomaly taxonomy, and (5) the evidence standard for root-level trace claims. Once those are fixed, each AI can independently propose a test sequence and predicted failure modes, and the results can be reconciled into a formal protocol.

Provisional doctrine

The black screen is not the closed window. The closed window is the boundary for which termination has been independently verified from both sides, residual state has been reconciled, and no alternate carrier or privileged dependency continues the relationship.

Appendix A — AI Response Review

ChatGPT: Strongest on separating display state from connection state, carrier switching, interface airlock/admission control, and incomplete-ontology framing. Recommended to retain these as core architecture.

Claude: Strongest on concise doctrine language, awareness stratification, convenience/exposure framing, and simple scenario-card structure. Its “unattended aperture” formulation is useful for training.

Google Gemini: Strongest on vivid narrative examples and the intuitive “idle threat” framing. Some proposed countermeasures are too mechanism-specific for a theory-agnostic first lab and should be replaced with logical containment controls.

Grok AI: Strongest on nested simulation stack, configurable Window Object properties, explicit infection-vector taxonomy, and tunable scenario variables. Recommended to retain the layered architecture and tunability.

Microsoft Copilot: No usable contribution was available for this iteration; do not infer agreement or disagreement from absence.

Appendix B — Minimal Data Schema

Field	Purpose
event_id	Unique monotonic identifier assigned at oversight layer
native_environment_id	Environment that reports the event
native_timestamp	Timestamp exactly as reported locally
native_calendar_schema	Calendar/epoch/time-reference metadata
control_monotonic_time	Authoritative ordering reference
observer_id	Agent/device/sensor producing observation
interface_id	Window/interface involved
presentation_state	What user sees
connection_state	What telemetry reports
event_open	Opening marker
event_close_requested	Requested closure
event_close_acknowledged	Counterparty acknowledgement
event_close_verified	Independent verification
residual_state	Any post-closure state
carrier_type	Device/environment/agent/message/etc.
source_provenance	Known source and transformation history
confidence	Evidence quality/confidence score

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